

Research project
BIO3308

How to write a
research article

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BIO3308



BIO3309



Learning outcomes

Find out about *why* to write research articles

Find out *how* research articles should be structured

...and how your article will be marked

2. Visual abstract (20%)

Brief Introduction, Methods,
Results, Discussion

Deadline: **9th February**

1. Research plan (20%)

Introduction, Methods,
Ethics

Deadline: **30th October**



3. Draft article (formative = NO mark)

Full research article draft:

- Abstract
- Lay reader abstract
- Introduction
- Methods
- Results
- Discussion
- References
- Appendix

Deadline: **9th March**

4. Research article (50%)

Deadline: **22nd April**

5. Supervisor's assessment (10%) of
Your research and transferable skills

Why writing research
articles?

Why writing research articles?

Advancement of science depends on clearly written scientific articles

Research that isn't communicated doesn't exist

Research that is not read or understood, doesn't make any difference to either science or society

Good academic writing has to be clear, accurate, professional and concise

Good writing is a key transferable skill

What makes a well written paper

ABSTRACT

A succinct,
accurate
summary
of the key
points
across all
sections

Introduction: clear
question & background,
why important

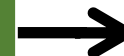
Aims & objectives /
hypothesis & predictions
clearly stated

Methods: all necessary
information for replication
(avoid unnecessary details)

Results: text *describes*
findings, useful figures
and tables

Discussion: explains the *meaning*
of the results in relation to the
question, how they results
advance the field, why important

Clear take home
message



Overview: before you start

What's your main goal/question?

How did you answer the question in a nutshell?

What is your key finding?

What's your main conclusion?

What's the major advance of your research?

What's the take home message?

Write 1-2 sentences for each = the skeleton of your paper

Visual Abstract



- It provides the essence of your research:
 - Main research question & aims/hypotheses & predictions
 - Main methods
 - Key findings
 - Major conclusions
 - Why it matters
 - Take home message
- The **skeleton of your own Research Article**, your 'story'

What is the difference
between scientific and
lay reader abstract?

Professional abstract

- Your elevator pitch to the **professional reader**, i.e. other scientists within or outside your field
- 1/2 sentence per section
 - Background and objective
 - A general and a specific question
 - Design
 - Results
 - Conclusion (take home message)
- NO stats
- NO references

Layman abstract

- Aimed at the **general public** and the **media**
- Similar summary but in plain language
- Fewer (no) details on methods
- NO stats
- NO references
- **Next lecture**

Introduction: from the bigger picture to the specific

Why should we care?

- Set the scene: describe the **general area of research**
- Highlight the relevance of this topic to the **big picture**

What do we know?

- **Critically review** the literature (**theory & methods**)
- **Critically evaluate key studies** relevant to the project

What do we not know?

- Highlight **gaps in knowledge**

What will your study do?

- Explain the **aims** of your project
- Explain your **hypothesis and predictions** or research question and aims

Research Plan

- This gives the structure of the Introduction & main Methods:
 - Background
 - Aims and objectives Hypothesis & predictions
 - Main methods
 - Ethics
- **Revise** for the Introduction following Supervisor's feedback
- but **mind self-plagiarism** (see **Research integrity** lecture)
- Your Research Plan is not the final Introduction!

Methods: the recipe

- Sufficient details to allow replication:
 - Clear and comprehensive but concise
- How was the evidence obtained?
 - Model organism/system and study site (if any)
 - Design: experiment, field, lab, bioinformatics...
 - Reagents, pipeline etc if appropriate
 - How were the data collected (if any)?
 - How were the data analysed?
 - The statistical approach (tests used, software & packages, etc)
- What did you do and **for what purpose?**
- Ethics statement if needed

Results

- What did you find?
- Be informative when describing your findings. E.g.:
 - Little informative: *There is a significant difference in height between men and women*
 - Highly informative: *Men are significantly taller than women*
- Write *non-significant* NOT *insignificant*
- All figures and tables must be cited in the text
- Don't break tables across pages
- Keep tables/figures and legend in the same page

How to present the Results

Logical structure

- e.g. by analysis/experiment matching the research questions or the objectives or the predictions

What to include:

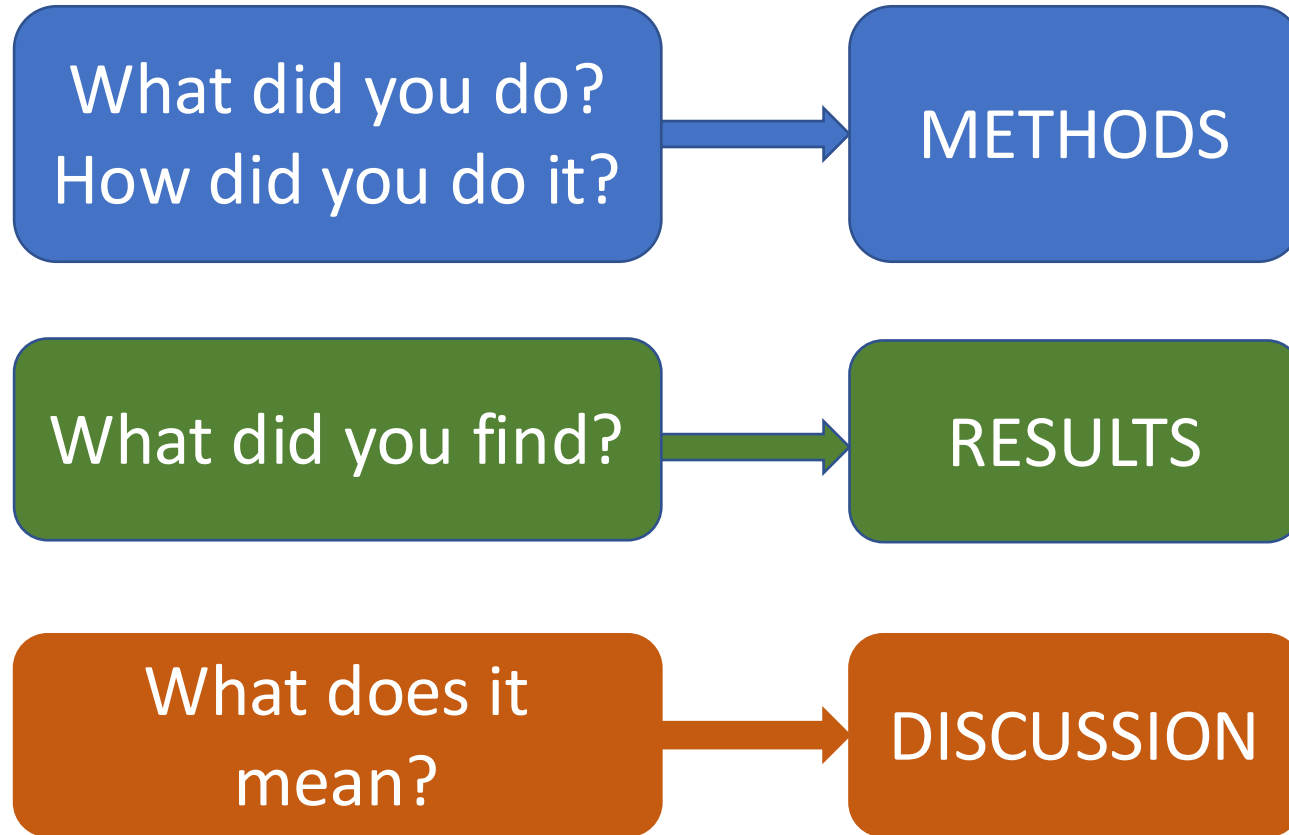
- **START** the section with text *describing* your findings
- Figures and/or Tables of data or analysis output
- Stats/modelling/any analytical output following academic conventions

Results

Lecture on *Presenting results* in oral and written format:

- General guidance on structure
- Text vs tables vs figure
- Checklist for good tables
- Checklist for good figures

What to include where?



IMPORTANT - in the **RESULTS** section include:

- Only findings
- NOT methods
- NOT interpretation
- NOT comparisons with works by others

Discussion: from the specific to the bigger picture

What do your results mean?

- **FIRST paragraph:** remind (but do not re-explain) the question/aim => **key** results & conclusion
- **NEXT: discuss** in depth **all** results: do they support the predictions? If yes, what does this mean? If no, what could be the reason?

How do your results compare to others?

- **Compare & contrast** with previous studies: similarities? Differences? Critique!
- Can you draw **general conclusions**? Any reasons for differences?

What next?

- **What could be improved** in the research you did?
- What should the **next step** be in this line of research?

Why should we care?

- **Implications:** how do your results **advance the science** in your broader research area? Why is this **important**?
- **FINAL paragraph:** sum up main points, provide **take home message**

Results vs Discussion

RESULTS

- *Describe* the results in words
- Statistics/Modelling/Analytical output
- (Most) Figures & Tables

DISCUSSION

- *Explain* the results
- *Interpret* the results in relation to the hypothesis & predictions or aims & objectives of the study
- *Place* the results in the context of the wider literature

WARNING: Common mistakes

INTRODUCTION

- Does not explain **why** the project matters
- Or what it **aims** to do
- No hypothesis & predictions / aims & objectives
- Lack of critical thinking

METHODS

- Insufficient information to replicate the results
- Too much detail

RESULTS

- Start with figures or tables instead of text
- Explain methods or introduces new methods
- Discuss/Explain the results

DISCUSSION

- *Describes* the results OR *incorrect* conclusions
- Reports the stats, additional analyses
- *Does not explain* the meaning of the results
- *Does not explain* how the results advance science
- *No* take home message
- Lack of critical thinking

Lecture on critical thinking and writing

beyond describing facts and ideas, critical thinking involves:

Critical synthesis

- Gathering and *summarizing* relevant evidence from appropriate sources
- *Exploring* and *comparing* ideas and evidence



Critical analysis

- How?
- Why?

Explaining the evidence



Critical

evaluation

- What if?
- So what?
- What next?

Proposing new ideas

References

- Citations from the Peer Reviewed literature = reliable evidence!
- CITE the relevant literature: give credit where credit is due
- Missing key papers = sloppy science, superficial understanding of the field
- **References:**
 - Everything that is cited in the article must be included in the Reference list
 - All papers in the Reference list must be cited in the text

What makes clear writing?

Comprehension is affected by:

- complexity of the material
- length of words
- length of sentences
- length of paragraphs

What makes 'clarity' in writing?



Each paragraph starts with a **topical sentence** on the main point of the paragraph



Paragraph structure: **point-evidence-explain**



Paragraphs of about **150 words**



Paragraphs **logically follow** one another in a logical structure



Always check the **logic** of your statements, your paragraph and the whole paper

What makes 'clarity' in writing?



Sentence structure: subject → verb → object



Sentences are **short** (15-20 words, max 50) and each convey only 1 point



Sentences are **straight to the point** & avoid unnecessary words



Don't be wordy, convoluted, superficial



Remove empty fillers (E.g. 'it is interesting to note that' => Note that... OR Interestingly,...)

Empty fillers	
It would thus appear that	
It is considered that	
It is this that explains	
It is possible that the cause is	
In light of the fact that	
It is often the case that	
It is interesting to note that	
It is not impossible that	
A not unlikely cause could be that	
It seems that there can be little doubt that	

Empty fillers	Shorter equivalent
It would thus appear that	Apparently
It is considered that	[delete all]
It is this that explains	This explains
It is possible that the cause is	The cause may be
In light of the fact that	Because
It is often the case that	Often
It is interesting to note that	Interestingly,
It is not impossible that	It is possible that
A not unlikely cause could be that	A possible cause may be
It seems that there can be little doubt that	Therefore

Preparing a final draft

- Successful writing depends on progressive refinement
- **Read and re-read** slowly and critically your text
 - coverage: did you fulfil your stated objectives?
 - coherence: do your ideas hang together?
 - presentation: does your work have a professional appearance (grammar, spelling, clarity)
- **Proof-read for spelling & grammar**

Conforming to the guidelines

see Canvas:
Modules/
Assessment
details/
Research article

- **Cover page:** student number, project title, supervisor, declaration of own work and HOW you used AI, SDGs symbols
- **Abstract:** 300 words
- **Layman Summary:** 300 words
- **Beneficiaries of research:** up to 200 words
 - Identify the main beneficiaries of your research project and justify your statement
 - Check the Lectures on **Societal Impact (basic and applied science)** and on **Entrepreneurship**

Conforming to the guidelines

see Canvas:
Modules/
Assessment
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Research article

- **Article:** 4000 words. Subheadings must identify Introduction, Methods (including ethical statement and ethical approval number if any), Results, Discussion, References
- **Tables / Figures:** a maximum of 4 between them but can be multi-panel / Multi-table
- **References:** no maximum but a minimum of 40
- **Appendices (Optional):** R codes, gene bank numbers, Supplementary Figures and Tables BUT **NO Supplementary text for any section**

Conforming
to the
guidelines

see Canvas:
Modules/
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Research article

FORMATTING

- Text, figure and table legends, appendix: Arial 11, double-spaced.
- Tables: Arial 11, single spaced.
- Margins: 2 cm left, right, top and bottom
- Page numbers must be included.
- Word count by section (Abstract, Lay summary, Beneficiaries, Article) must be included.
- Submit as PDF

IMPORTANT: use of AI or other means

- AI tools may help summarise information
- **Cons:** cannot critically analyse the literature, may get facts wrong, invent or incorrectly cite the literature
- **Use of AI tools may breach academic integrity => misconduct**

QUB Academic Offences:

2.7 Contract cheating: where a student commissions or seek to commission (either paid or unpaid) another individual ***or artificial intelligence software tool*** to complete academic work on their behalf

IMPORTANT: use of AI or other means

- You now must include in your declaration in the cover page of your article submission:
- *I certify that the submission is my own work, all sources are correctly attributed and the contribution of any assistive technologies is fully acknowledged.*
- *AI tools have **[not]** been used in **[state how you used it]** as agreed with the Supervisor at the start of the project.*

1. Research plan (20%)
Deadline: **30th October**

Feedback

2. Visual abstract (20%)
Deadline: **9th February**

Feedback

**3. Draft article
(formative = NO mark)**
Deadline: **9th March**

Feedback

4. Research article (50%)
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Feedback



Support sessions



Content

Introduction, **Supervisor's assessment, Research Plan**

Research integrity; Bioethics

Experimental design; Critical thinking

Systematic reviews & meta-analyses; Record keeping

Literature Search & Referencing

Beneficiaries of research, entrepreneurship

Using feedback; Presenting results of quantitative studies

Qualitative methods, pedagogical research

Assessment 2: Visual abstract

Titles and scientific abstract

Assessment 3 &4: Research article

Science communication and lay abstract

Mark scheme: for each section & writing style

High first - An exceptionally **clear, correct and accurate... Exceptional** level of **critical analysis and evaluation** of the literature ...demonstrating **excellent understanding ...exceptionally well explained... logical** and well structured

<u>High 1st</u> Exceptional Exemplary	<u>Low 1st</u> Excellent	<u>2.1</u> Very good Few minor omissions or errors	<u>2.2</u> Good Some omissions or errors	<u>3rd</u> Adequate Sufficient Few major errors or omissions	<u>Fail</u> Insufficient Muddled Many major errors or omissions
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Marking rubric available: in Canvas BIO3308 / Assignments / Research Article Draft & Research Article

Submission Research article draft

- **Only feedback and only on 1 draft**
- **SUBMIT A COMPLETE DRAFT**
- Students that do not submit a complete draft typically do less well in the final submission
- Formative: no mark
- Same marking criteria, same formatting guidelines as the Final Article
- You can submit before the deadline (coordinate with your supervisor)
- Discussion board in Canvas for generic queries

Submission: Final research article

- **2 identical submissions:**
 - Supervisor & Second marker
- Be independent: revise while waiting for feedback on the draft
- Peer help
- **Start writing now!**
- **VERY IMPORTANT:** give your data and lab books/diary to your supervisor
- Next and final session 3rd March: Science communication and lay reader abstract